AN ENGINEERING DECISION

A queue is not extra capacity.

Buffer the burst. Budget the recovery.

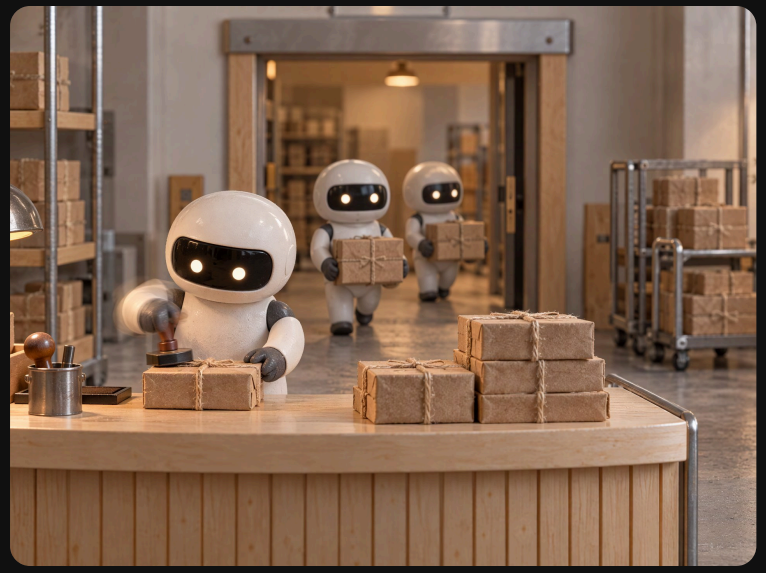


01

Capacity and waiting

THE ILLUSION

Fast acceptance.
Unchanged capacity.



- Accepted in milliseconds.
- Finished at the same rate.

A waiting room.
Not a wider exit.



RECOVERY

Only spare capacity drains the backlog.



Completions must outrun new arrivals.

Full utilization leaves nothing for recovery.

02

The hidden cost of acceptance

Watch the oldest unfinished promise.



- Queue depth counts work.
- Job age exposes delay.
- Include in-flight jobs.

Retries are more arrivals.

- One identity per logical job.
- One finite retry budget.



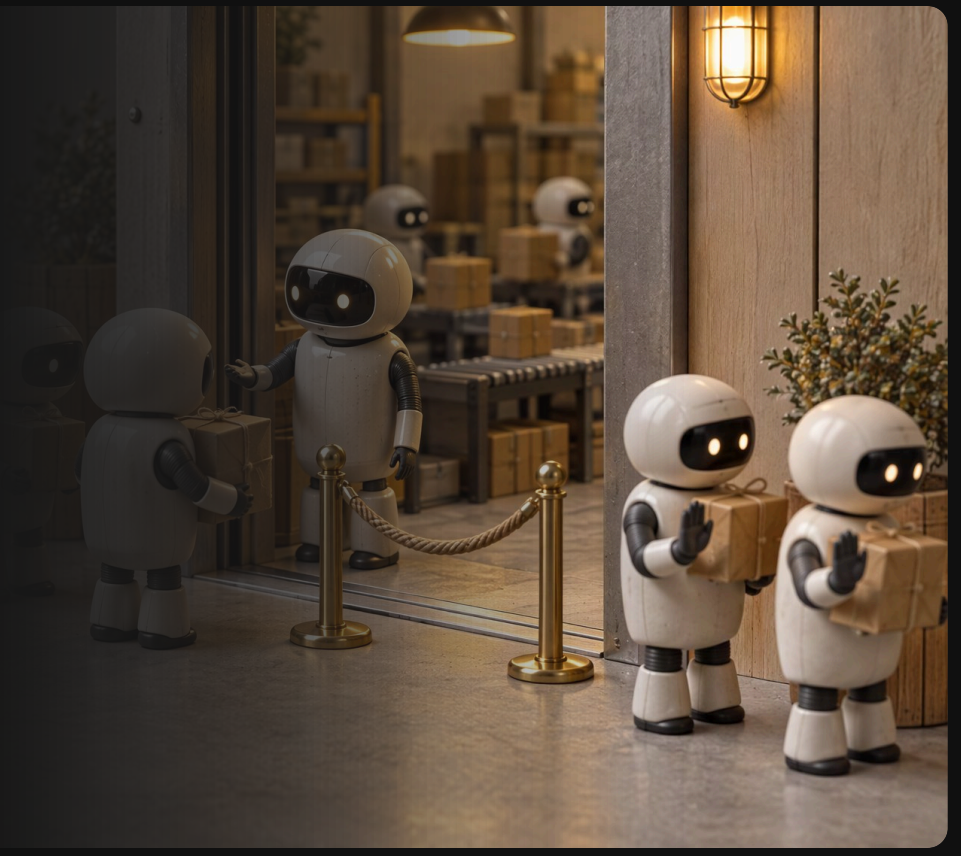
03

A bounded promise

ADMISSION

Bound the promise, not just the queue.

- Useful deadline.
- Hard storage limit.
- Honest rejection.



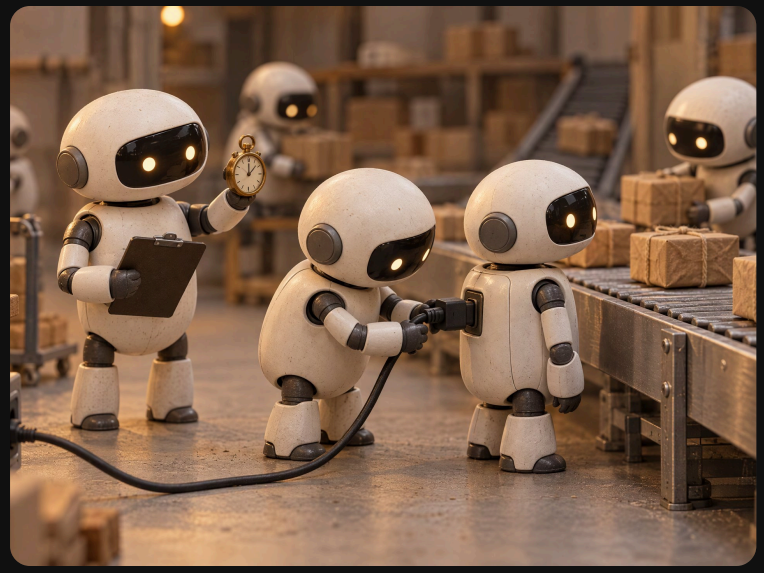
Every job needs
a visible ending.



Completed. Failed, with an owner. Explicitly expired.
Never just waiting.

THE DECISION TEST

Prove recovery under continuing load.



- Burst. Slow dependency. Worker restart.
- Account for every accepted job.

THE KEY MESSAGE

Buffer the burst.
Budget the recovery.

A queue buys time, not capacity.

